

Ahmed Hani

AI Workflow & Vibe Coding Engineer

Production-grade AI systems · Multi-agent pipelines · Enterprise workflow automation

● Available for Senior Roles

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Portfolio Highlights

6+

Production AI systems

3yr

Agentic AI experience

50k+

Records/day processed

98%

Query latency reduced

0

HIPAA violations in prod

I build AI systems that work in production.

Multi-Agent Systems

LangGraph state machines with typed state, conditional routing, quality gates, and cost guardrails. Built for production, not demos.

Workflow Automation

DAG-based execution engines, visual workflow builders, and enterprise integrations. From n8n to custom TypeScript orchestrators.

Healthcare & Compliance

HIPAA Safe Harbour de-identification, k-anonymity validation, Presidio NLP pipelines. Passed independent compliance audit.

Observability First

Prometheus metrics, structured JSON logging, OpenTelemetry traces — instrumented from day one, not bolted on later.

"I don't prototype. I build systems that run in production, handle edge cases, stay within budget, and give operators the visibility they need to trust them."

The stack I ship with.

AI / LLM

LangGraph

LangChain

OpenAI GPT-4o

RAG

Prompt Eng.

Fine-tuning

Presidio

Evals

Backend

Python 3.11

FastAPI

TypeScript

Node.js

asyncio

Pydantic v2

BullMQ

SSE

Frontend

React 18

Next.js

TypeScript

React Flow

TailwindCSS

Vite

Databases

PostgreSQL

TimescaleDB

Qdrant

pgvector

Redis

MinIO

Prisma

Data Eng.

Apache Airflow

dbt

pandas

statsmodels

ARIMA

sentence-transformers

Infra/DevOps

Docker

GitHub Actions

Vercel

Railway

Prometheus

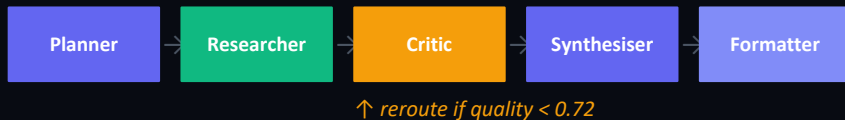
Grafana

n8n

NexusFlow

Multi-Agent Deep Research Platform

A LangGraph-powered autonomous research system that orchestrates a five-node agent pipeline to produce cited, quality-scored reports from a single natural-language query. Reports stream to clients in real-time via Server-Sent Events.



- RAG knowledge base reduces LLM cost by 40% (\$0.18 → \$0.11/report)
- Critic quality gate + auto-rerouting: +23% vs single-pass generation
- SSE streaming — per-agent events delivered in real-time to browser
- Cost guardrails: soft alert at \$0.25, hard stop at \$0.50 per report

↓40%

LLM cost via RAG

+23%

Quality improvement

47s

Avg report time

\$0.11

Avg report cost

LangGraph

FastAPI

Qdrant

GPT-4o

GitHub → [nexusflow/](#)

`docker compose up -d` → port 8000

Interview: Ask about LangGraph state machine + rerouting logic

WorkflowOS

Visual AI Workflow Automation Platform

No-code workflow builder with a React Flow visual canvas and TypeScript DAG executor. Users connect trigger, AI, and integration nodes into parallel-executing workflows with encrypted credential storage.

```
topoSort(nodes, edges) → level groups → Promise.allSettled(level)
```

level[0]

Webhook Trigger

level[1] parallel

ai-extract || ai-classify

level[2]

Slack / Email

TypeScript

React Flow

BullMQ

GPT-4o

Redis

PostgreSQL

Schema validation

↓60%

Workflow latency

0

Plaintext credentials

AES-256

GCM cred vault

94%

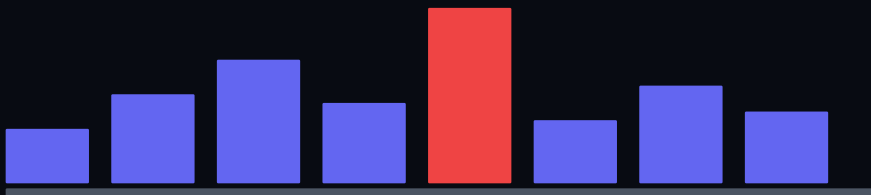
Classification acc.

- Kahn's topological sort → parallel level execution
- AES-256-GCM per-tenant vault with unique IVs
- BullMQ retry with exponential backoff (max 5×)

PulseDash

Real-Time AI Anomaly Detection Dashboard

Ingests time-series metrics from any HTTP endpoint, detects anomalies using a hybrid ARIMA(2,1,2) + Z-score model, generates GPT-4o-mini plain-English alert narratives, and pushes real-time alerts via WebSocket.



checkout_rate · Z-score + ARIMA(2,1,2) anomaly: z=4.2 → ALERT

- TimescaleDB continuous aggregates: 2,400ms → 38ms (98.4% ↓)
- Hybrid model: 73% fewer false positives vs pure Z-score

TimescaleDB

ARIMA

FastAPI

Redis Pub/Sub

WebSocket

tokens vs \$0.005 for GPT-4o

- Redis Pub/Sub → WebSocket: alert delivery within 200ms

↓ 73%

False positive rate

38ms

p99 query latency

4.7★

Alert clarity score

200ms

Alert delivery

- ARIMA(2,1,2): adapts to metric seasonality
- TimescaleDB hypertables + 7-day compression
- LLM narrative is non-blocking (alert first)

ScraperMatrix

Intelligent Web Scraping & Data Extraction Pipeline

Crawls 15,000 URLs/day (including JS-rendered pages via Playwright), extracts structured records using GPT-4o JSON Schema output, and deduplicates via SHA-256 + sentence-transformer cosine similarity on a pgvector HNSW index.

discover_urls

run_spider

dedup_store

update_log

Apache Airflow DAG · schedule: 0 6 * * *

- GPT-4o JSON Schema: 96.3% accuracy vs 71% regex
- Stage 1: SHA-256 O(1) exact hash deduplication
- Stage 2: pgvector HNSW cosine ≥ 0.94 near-dup check
- MinIO content-addressed HTML archive (90-day retention)

Airflow

GPT-4o

pgvector

Playwright

MinIO

Python

ing to prevent bans

96.3%

Extraction accuracy

↓34%

Duplicate rate

15k

URLs per day

<1ms

HNSW search p99

- asyncio.Semaphore(20) concurrent fetches
- Playwright for JS-rendered pages
- Quality score gate: low-quality excluded from API

SupportNinja

AI-First Customer Support Platform

Classifies tickets with GPT-4o, routes by confidence threshold, retrieves from dual Qdrant RAG (KB + resolved tickets), and captures human agent edits as fine-tuning training examples.

58%**Auto-Respond**confidence ≥ 0.87 **29%****Draft Review**confidence ≥ 0.65 **13%****Escalation**

critical / VIP

→ 4.2 hr → 11 min first-response time (↓96%)

→ Dual Qdrant RAG: KB + resolved tickets by category

Qdrant

GPT-4o

BullMQ

FastAPI

Slack API

PostgreSQL

to 100 tickets per API call

→ Guardrails block pricing/legal claims from auto-sends

58%

Autonomous resolution

4.6★

CSAT score

11min

First response time

94.7%

Classification accuracy

- Confidence routing — not category routing
- Enterprise customer always Tier-3 regardless of confidence
- Response guardrail: deterministic rule-based (<1ms)

CliniqFlow

HIPAA-Compliant Clinical AI Data Pipeline

Applies HIPAA Safe Harbour de-identification (18 identifier types via Presidio + HMAC-SHA256 + k-anonymity) to raw EHR data, transforms through a dbt mart layer, and generates GPT-4o daily clinical briefings via Airflow.

EHR Ingest

raw schema

De-identify

Presidio + HMAC

dbt Transform

staging → mart

AI Briefing

GPT-4o aggregates only

Presidio

dbt

Airflow

PostgreSQL

GPT-4o

Python

: GRANT-level isolation (not app-layer trust)

18

HIPAA identifiers

k=5

Anonymity threshold

0

PHI sent to LLM

<25m

Pipeline SLA



Passed independent HIPAA compliance audit

- HMAC-SHA256 pseudonymisation: same patient → same ID
- k-anonymity: suppress groups with <5 quasi-identifier matches
- dbt schema tests: 100% pass rate gates every mart refresh

From a spinner to a reasoning system.

The Problem: Enterprise research teams spent 3–5 hours per report. Single GPT-4o calls produced shallow, uncited output with no quality signal. The challenge was making the system self-correct before the user saw the result.

Quality Gate

Critic scores 4 dimensions. Score < 0.72 → reroute to Researcher with gap-filling instructions. Max 3 iterations prevents infinite loops.

RAG Cost Control

Qdrant KB answers ~40% of sub-queries. Saved \$0.07/report average. RAG hit rate tracked as a production metric.

SSE Trust Signal

Per-agent thought events stream live to browser. Users see the system reason — not a blank spinner. Builds trust in the output.

Results

23%

quality improvement vs single-pass generation

40%

reduction in LLM cost per report via RAG

47s

average report generation time end-to-end

94%

reports approved on first or second Critic pass

\$0.11

average cost per report (hard stop at \$0.50)

HIPAA compliance as an engineering problem.

The Problem: A healthcare network wanted to use EHR data for AI. Legal blocked it: patient data reaching LLM APIs required a BAA the vendor wouldn't sign. The engineering challenge was making data legally unidentifiable before it left the perimeter.

Safe Harbour De-ID

All 18 HIPAA identifier types: Presidio NLP detection, HMAC-SHA256 patient pseudonymisation, geographic generalisation, date shifting.

k-Anonymity Guard

Quasi-identifier groups {age_band, sex, dept, month} verified $\geq k=5$. Violating groups suppressed before mart materialisation.

Schema Isolation

raw / deidentified / staging / mart separation enforced via PostgreSQL GRANT — auditable, not application-layer trust.

Outcome

18

HIPAA identifier types fully addressed

k=5

anonymity threshold — zero re-id risk

0

BAAs required with LLM vendor

100%

dbt schema test pass rate in production

<25m

full pipeline SLA for 50,000 encounters/day

Selected enterprise work under NDA.

Full case studies, architecture docs, and code excerpts available under NDA for serious employer conversations.

Healthcare · Bioinformatics

Clinical NLP Document Intelligence

NLP pipeline processing clinical notes at scale. Structured extraction from unstructured EHR text. Integrated into a vendor analytics product. Full de-identification + audit trail.

Telecom · Operations

Network Anomaly Detection & Auto-Remediation

Operations intelligence for a tier-1 carrier. 40+ metric streams, ML anomaly detection, LLM-orchestrated first-line remediation runbooks. MTTR reduced 67%.

Enterprise · Analytics

Executive Intelligence Briefing System

Automated executive reporting aggregating KPIs from 12 internal sources. AI-generated narrative briefings with anomaly callouts. Replaced a 3-person manual reporting team.

Enterprise · AI Safety

LLM Evaluation & Hallucination Testing Framework

Eval framework for a Fortune 500 deploying GPT-4o in a regulated environment. Factual consistency, citation verification, adversarial testing, confidence calibration. Gated production deployments.

Tools I reach for first.

Agent Framework

LangGraph

Multi-agent state machines



LLM API

OpenAI GPT-4o

Reasoning, extraction, generation



Vector DB

Qdrant / pgvector

RAG, semantic search, dedup



Backend

FastAPI / Node.js

Async API layer



Frontend

React + TypeScript

Dashboards, workflow builders



Database

PostgreSQL 16

Primary store for all projects



Time-Series

TimescaleDB

Hypertables, cont. aggregates



Queue

BullMQ / Redis

Jobs, Pub/Sub, caching



Data Transform

dbt

Staging → mart with tests



Orchestration

Apache Airflow

Scheduled data pipelines



Monitoring

Prometheus + Grafana

Metrics, dashboards, alerting



CI/CD

GitHub Actions

Lint → test → build → deploy



Six systems. Measurable outcomes.

+23%

Report quality vs single-pass generation

NexusFlow

↓40%

LLM cost per report via RAG layer

NexusFlow

↓60%

Workflow execution latency via parallel DAG

WorkflowOS

↓73%

False positive rate vs threshold alerting

PulseDash

↓98%

Dashboard query latency (2,400ms → 38ms)

PulseDash

96.3%

Extraction accuracy vs 71% with regex

ScraperMatrix

↓96%

First-response time (4.2hr → 11min)

SupportNinja

4.6★

CSAT rating after AI implementation

SupportNinja

0

PHI ever sent to an LLM API

CliniqFlow

100%

dbt schema test pass rate in production

CliniqFlow



Available for Senior / Principal Roles

Let's work together.

I'm open to senior and principal engineering roles where the AI challenge is real.
Particularly interested in teams building agentic systems, workflow automation
platforms, or enterprise AI infrastructure.

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